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Anthony A. Snead

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EDUCATION

Ph.D., Biology: University of Alabama **May 2023**
Dissertation: Environmental Responses Across Biological Scales: Abiotic Drivers of Species Range, Population Size, and Genetic Variation in the Mangrove Rivulus, *Kryptolebias marmoratus*
Advisor: Ryan L Earley

B.S., Environmental Biology: University of South Florida **2018**

APPOINTMENTS

NSF Postdoctoral Research Fellow **June 2023 - Present**
Kristin M Winchell, Department of Biology, New York University

RESEARCH PROGRAM

Evolutionary responses to novelty are shaped—and often constrained—by genome architecture. My research defines how. I study how genome structure canalizes adaptive evolution and shapes evolutionary trajectories. Grounded in evolutionary biology, my research moves across scales to link evolutionary history to phenotypic change by uncovering the molecular mechanisms through which genome architecture shapes adaptation. I test how structural variants and single nucleotide polymorphisms generate phenotypic diversity through distinct molecular mechanisms, clarifying how genome architecture shapes both static and plastic trait evolution. Working in ecologically relevant, non-model systems experiencing environmental novelty, I develop predictive frameworks that uncover how genome architecture dictates evolutionary shifts to uncover the molecular mechanisms that generate phenotypic diversity.

PUBLICATION

(† shared first-authorship, * undergraduate author)

ORCID: 0000-0002-5020-8729 *Google Scholar ID: TbSh_2oAAAAJ&hl*

Submitted

20. †Kauffman KM, †Daniels J, †**Snead AA**, †Payne SAR (202X). Professional Biologist discusses issues in the practice of biological professions. *Under Revision*: *Bioscience*

Published

19. Alaasam V, **Snead AA**, Thonis A, Winchell KM (2026). Eco-evolutionary dynamics shaping biodiversity in the urban mosaic. *Nat Rev Biodivers*. (doi: 10.1038/s44358-026-00138-0)
18. Fang M, **Snead AA**, *Zhang AY, Munshi-South J, Winchell KM. (2026) Cities as evolutionary incubators for the global spread of the spotted lanternfly. *Proc Biol Sci* 293(2064): 20252292. (doi: 10.1101/2025.06.30.662460).
17. Cunha Neto IL, **Snead AA**, Landis JB, Specht CD, Onyenedum JG (2026). Ectopic cambia in wisteria vines are associated with the expression of conserved KNOX genes. *Nat Comm* (doi: 10.1038/s41467-026-68669-w).

Summary: In this work, we identify a conserved gene regulatory mechanism underlying the formation of ectopic vascular tissues. By integrating comparative developmental analyses with gene expression profiling, we show that altered expression of KNOX transcription factors is associated with novel cambial activity and morphological innovation. These results directly link regulatory gene expression to developmental patterning and complex trait formation.

16. de Greef E, Müller C, **Snead AA**, Rivkin LR, Ferguson SH, Watt CA, Marcoux M, Peterson SD, Garroway CJ (202X). Identifying areas of potential risk based on future genetic adaptability in three Arctic whale species. *Am Nat* (doi: 10.1086/738889)
15. **Snead AA**, *Lin E, *Linsky S, *Olson L, Winchell KM (2025). Conserved Ecological Responses to Novel Urban Stressors. *J Biogeogr* 52(12), e70104 (doi: 10.111/jbi.70104)
14. **Snead AA**, Alaasam V, Winchell KM (2025). Toward a Cross -Disciplinary Understanding of Urban Evolution. *Integr Comp Biol* 65(2), 229-235 (doi: 10.1093/icb/icaf117).
13. Stahl L, **Snead AA**, Olson J (2025). Predicting Naegleria fowleri Freshwater Habitat Suitability in the United States for the Present-Day and Future using Ecological Niche Modeling. *Water Res* 124018 (doi: 10.1016/j.watres.2025.124018).
12. **Snead AA**, *Quackenbush CR, *Trojahn S, *McDonald AL, Lins L, *Cornelius C, Adams PE, Ma D, Hsu Y, Haag E, Silvestre F, Kanamori A, Earley RL, Kelley JL (2025). Embryonic thermal environments drive plasticity in gene expression. *Fish Physiol Biochem* 51(111) (doi: 10.1007/s10695-025-01522-x).

Summary: This study demonstrates how early developmental environments induce coordinated and reproducible changes in gene expression, revealing regulatory mechanisms underlying phenotypic plasticity. Using RNA-seq, we identify transcriptional programs responsive to thermal conditions during embryogenesis and link these expression shifts to downstream phenotypic outcomes. This work establishes a mechanistic connection between environmental input, transcriptional regulation, and complex trait variation.

11. Torres-Sanchez et al. [including **Snead AA**] (2025). Comparative biology needs de novo transcriptome assemblies: drawing attention towards amphibians. *AMRE* 46(3), 283-30 (doi: 10.1163/15685381-bja10232).

Summary: This study highlights the importance of high-quality transcriptome assemblies for functional and comparative genomics. We demonstrate how de novo transcriptomic resources enable robust analyses of gene expression, regulatory evolution, and molecular mechanisms in non-model systems, providing critical infrastructure for mechanistic studies of complex traits.

10. †**Snead AA**, †Meng F, *Largotta N, Winchell KM, Levine BA (2025). Diploid chromosome-level genome assembly for *Lycorma delicatula Sci Data* 12(579) (doi: 10.1038/s41597-025-04854-8).

Summary: I led the generation and annotation of the first diploid, chromosome-level genome assembly for *L. delicatula*, providing a high-resolution framework for studying genome architecture and regulatory variation. This resource enables the characterization of structural variants, regulatory elements, and gene expression underlying adaptive and complex traits, and supports downstream functional and comparative analyses.

9. Kosch TA et al. [including **Snead AA**] (2024). The Amphibian Genomics Consortium: advancing genomic and genetic resources for amphibian research and conservation. *BMC Genomics* 25(1025) (doi: 10.1186/s12864-024-10899-7).
8. Gustafson JA et al. [including **Snead AA**] (2024). High-coverage nanopore sequencing of samples from the 1000 Genomes Project to build a comprehensive catalog of human genetic variation. *Genome Res*. 34(11), 2061-2073 (doi: 10.1101/gr.279273.124).
7. **Snead AA**, Tatarenkov A, Taylor DS, Marson K, Earley RL (2024). Centrality to the Metapopulation is more Important for Population Genetic Diversity than Habitat Area or Fragmentation. *Biology Letters* 20(7): 20240158 (doi: 10.1098/rsbl.2024.0158).
6. **Snead AA**, Tatarenkov A, Avise JC, Taylor DS, Turner BJ, Marson K, Earley RL (2023). Out to sea: Ocean currents and patterns of asymmetric gene flow in an intertidal fish species. *Front Genet* 14: 1206543 (doi: 10.3389/fgene.2023.1206543).
5. Clark RD, Fitz KS, Garcia E, Jaynes KE, Reid BN, Sawkins A, **Snead AA**, Whalen J, Pinsky ML (2023) The Practice and Promise of Temporal Genomics for Measuring Evolutionary Responses to Global Change. *Mol Ecol Res* 25(5): e13789 (doi: 10.1111/1755-0998.13789).

4. **Snead AA** & Clark RD (2022). The Biological Hierarchy, Time, and Temporal 'Omics in Evolutionary Biology: A Perspective. *Integr Comp Biol* 62(6), 1872-1886 (doi: 10.1093/icb/icac146).
3. **Snead AA** & Alda F (2022). Time Series Sequences for Evolutionary Inferences. *Integr Comp Biol* 62(6), 1771-1783 (doi: 10.1093/icb/icac138).
2. **Snead AA** & Earley RL (2022). Predicting the in-Between: Present and future habitat suitability of an intertidal euryhaline fish. *Ecol Inform* (doi: 10.1016/j.ecoinf.2021.101523).
1. Davidson J, Summerfelt S, Espmark AMO, Mota V, Marancik D, Earley RL, **Snead AA**, Good C (2021). Effects of ozone on post-smolt Atlantic salmon (Salmo salar) performance, health, and maturation in freshwater recirculation aquaculture systems. *Aquac* 533:736208 (doi: 10.1016/j.aquaculture.2020.736208).

GRANTS & FELLOWSHIPS (**$339,829.32 Total**)

Under Review

National Institute of Health Pathway to Independence Award (K99/R00) **$997,000**
Mechanistic Links Between Structural Variation and Phenotypic Diversity in Natural Populations

Awarded - Accepted

National Science Foundation Postdoctoral Research Fellowship in Biology **$240,000**
Salamanders in the City: An Integrative Investigation into Microevolutionary Change

University of Alabama Aquatic Biology Enhancement Assistantship **$54,538**
2022-2023 Academic Year

RCN for Evolution in Changing Seas 2020 Call for Working Groups (Co-PI) **$15,762.50**
Implications of Methodology on Temporal Genomic Inferences

E.O. Wilson Biodiversity Fellowship Summer 2020 **$6,944**
Population Genetic and Environmental DNA Sampling in the Florida Keys

The College Academy of Research, Scholarship, and Creative Activity Grant **$5,999**
If You Can't See It, It Doesn't Exist. Or Does It? Validating a Technique to Estimate Abundance in a Vulnerable, Highly Enigmatic Marine Fish.

Office of Research and Economic Development Small Grant Program **$5,844**
Validating a Non-Invasive Method for Estimating Abundance in Threatened or Cryptic Marine Fishes.

SPRI Grant for Genomic Research (Beckman Coulter) **$3,000**
Eurycea bislineata: A model for studying salamander responses to environmental change

Company of Biologists' Scientific Meeting Grant (Co-PI) **$2,741.82**
Cities as Natural Experiments: Eco-Evolutionary Responses to Urbanization

Awarded - Declined

Presidential Management Fellowship (PMF) **~$150,000**
Office of Presidential Management

Small Grants

Sigma XI (**$1,000**), Summer Institute in Statistical Genetics (**$900**), Bishop-Stackman Marine Science Endowed Grant (**$500**), Travel Grants (NYU: **$700**; UA:**$500**; ESA: **$500**, **$400**)

SEMINARS, CONFERENCES & SYMPOSIA

Invited Presentations

University of Pennsylvania – Philadelphia, Pennsylvania, Biology Departmental Seminar **2025**

Villanova University – Philadelphia, Pennsylvania Philadelphia Herpetological Meeting **2024**

American Museum of Natural History – New York, New York, SDM Group **2024**

University of South Florida – Tampa, Florida, Departmental Seminar **2023**

Fresno State – Virtual, Departmental Seminar **2022**

Saint Leo University – Virtual, Departmental Seminar **2021**

Submitted Presentations (o – oral, p – poster)

[o]Society for Integrative and Comparative Biology Meeting – Portland, Oregon **2026**

[o]Society for Integrative and Comparative Biology Meeting – Atlanta, Georgia **2025**

[o]Society for Integrative and Comparative Biology Meeting – Atlanta, Georgia **2025**

[o]Society for Integrative and Comparative Biology Meeting – Seattle, Washington **2024**

[o]Ecological Society of America Meeting – Portland, Oregon **2023**

[p]Society for Molecular Biology and Evolution: Small Populations – Princeton, New Jersey **2023**

[o]Society for Integrative and Comparative Biology Meeting – Austin, Texas **2023**

[o]Ecological Society of America Meeting – Montreal, Canada **2022**

[o]Fisheries Society of the British Isles – Virtual **2022**

Organized – Symposia & Lecture Series

Symposium: Urban Evolution – Society for Integrative and Comparative Biology Meeting **2025**

Lecture Series: Temporal Genomics – Virtual **2021**

Organized – Workshops

[o]Society for Integrative and Comparative Biology Meeting – Atlanta, Georgia **2025**

TEACHING EXPERIENCE

Instructor of Record

Introduction to Principles of Biology (BSC 113) **2022**
- **UA Department of Biological Sciences –** 14 Lectures
- Spring 2022 – Enrollment: 34 (Section 1), 31 (Section 2)

Intro Biology II for Non-Majors Lab (BSC 109L) **2020 – 2021**
- **UA Department of Biological Sciences –** 14 Classes
- Fall 2020 – Enrollment: 26 (Section 1)
- Spring 2021 – Enrollment: 30 (Section 1), 28 (Section 2)
- Fall 2021 – Enrollment: 28 (Section 1)

Principles of Biology II Lab (BSC 117) **2018 – 2021**
- **UA Department of Biological Sciences –** 14 Classes
- Fall 2018 – Enrollment: 28 (Section 1), 30 (Section 2)
- Spring 2019 – Enrollment: 30 (Section 1), 30 (Section 2)
- Summer 2019 – Enrollment 14 (Section 1)
- Fall 2019 – Enrollment: 29 (Section 1), 28 (Section 2)
- Spring 2020 – Enrollment: 30 (Section 1)
- Fall 2020 – Enrollment: 30 (Section 1)
- Fall 2021 – Enrollment: 29 (Section 1)

Graduate Teaching Assistant

Research Seminars in Biology (BSC 488) **2021**
Spring – Enrollment: 61 (Section 1)

Biological Sciences Seminar (BSC 610) **2021**
Spring – Enrollment: 75 (Section 1)

SERVICE

Internal

UA Biological Sciences Diversity Equity, & Inclusion Committee **Fall 2021 – Summer 2023**

External

Squamata Genomics Consortium (Member) **2025 – Present**

Amphibian Genomics Consortium (Member) **2024 – Present**

eDNA Technology Readiness Level Group (Member) **2023 – 2024**

Clear Directions Mentoring (Mentorship Support Committee) **2023 – 2024**

1000 Genomes ONT Sequencing Consortium (Member) **2022 – Present**

RCN-ECS Reading Group – Temporal Data & Evolutionary Change (Organizer) **Spring 2022**

Editorial Service

Co-Organizer, Integrative & Comparative Biology, 2025. Special issue on "Urban Evolution."

Organizer/Associate Editor, Integrative & Comparative Biology, 2022. Special issue on "Temporal Genomics

Journal Reviewed for

Nature Communications, Ecological Informatics, Molecular Phylogenetics and Evolution, Ecological Processes, Aquaculture Research, Wellcome Open Research, Ecology and Evolution, Cell Reports

Conferences Reviewed for

International Biogeography Society, Ecological Society of America

Professional Membership

Society for Modeling & Theory in Population Biology (SMTPB), New York Academy of Sciences, Society for Molecular Biology & Evolution (SMBE), Ecological Society of America (ESA), The International Biogeography Society (IBS), Society for Integrative & Comparative Biology (SICB), Society for the Study of Evolution (SSE), Sigma Xi, American Association for the Advancement of Science

Outreach

Denver Metro Regional Science & Engineering Fair (*Scientific Review Board*) **2024**

Clear Direction Mentoring (*Mentor*) **2023 – 2024**

Denver Metro Regional Science & Engineering Fair (*Volunteer Judge*) **2022**

Meet a Bama Biology (*Scientific Volunteer*) **2021 – 2022**

UA Undergraduate Research $ Creative Activity Conference (*Volunteer Judge*) **2021, 2022**

UA STEM Showcase (*Volunteer Judge*) **2021**

Letters to a Pre-Scientist (*Scientific Volunteer*) **2020 – Present**

UA SafeZone (*Ally/Trainer*) **2018 – 2023**

The University of Alabama's Night at the Museum (*Volunteer Instructor*) **2019**

Skype a Scientist (*Scientific Volunteer*) **2018 – Present**

Media Coverage

Schmelzer H. 2022. Microevolution can guide conservation efforts. NatureVolve 11:26-29.
* Interview & Popular Science Article Spotlight

REFERENCES

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